

## CAPSTONE PROJECT – ASSESSING PROFESSOR EFFECTIVENESS

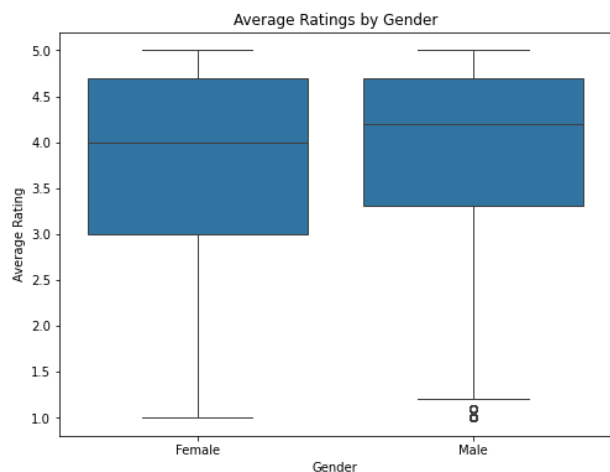
**Data cleaning and preprocessing:** Data preprocessing and cleaning were performed to ensure that the data was suitable for analysis. First, column headers were added to clearly identify the variables in the dataset. Then, considering that there were many observations where teachers had been rated by only one student (37522), only observations with more than one rating were accepted to avoid high skew and ensure that the average rating is more meaningful. Although this resulted in a significant loss of 41.7% of the original data, we accept this since we still have more than 50% of the dataset which is also more representative for assessing professor effectiveness. Lastly, the remaining missing values were addressed by imputing with the mean for continuous variables, median for variables with whole number values, and mode for categorical variables. Throughout the project, the necessity for Principal Component Analysis (PCA) was checked when working with multiple predictors but was not performed since correlation matrices revealed low correlation between the chosen features. The code for plotting will not be commented on as it is straightforward and uniform through the use of Matplotlib and can be easily understood through the plots themselves.

### 1. Is there evidence of pro-male gender bias in this dataset?

To assess pro-male gender bias, the following metrics were considered: average rating, average difficulty, retake proportion, and pepper (hotness). While average rating is our principal variable for consideration, other variables were also assessed for completeness of understanding. Then, Welch's t-test was performed for the above variables, with a boxplot graphed for average rating in specific.

**Results:** Considering an alpha level of 0.005, there is a significant bias for average rating (**p-value: 3.75e-09**) and retake proportion (**p-value: 0.0007**) with a positive t-statistic (**5.90** and **3.39** respectively) confirming a pro-male bias. The boxplot also confirms these findings with a noticeable difference in average ratings between genders. No significant bias was detected for average difficulty and pepper.

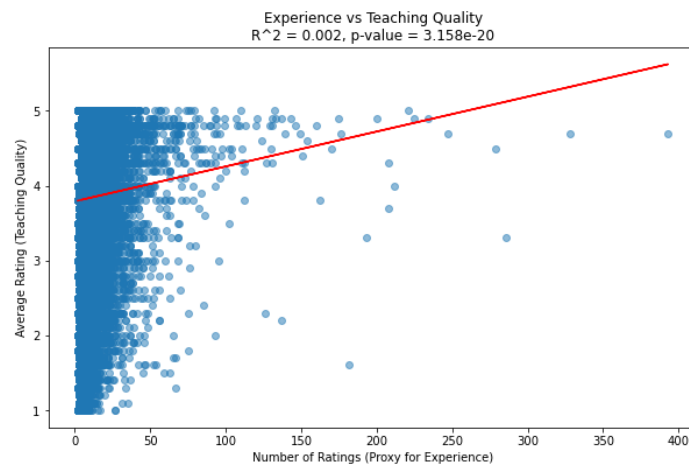
```
Gender Bias Results:
avg_rating:
T-Statistic: 5.895978806862923
P-Value: 3.753762277113109e-09
avg_difficulty:
T-Statistic: -0.2532688635202437
P-Value: 0.8000617106351611
retake_prop:
T-Statistic: 3.3856108519179924
P-Value: 0.000710860116522626
pepper:
T-Statistic: -2.127604780163336
P-Value: 0.03337575925157848
```



## 2. Is there an effect of experience on the quality of teaching?

To assess whether there is an effect of experience on the quality of teaching, a Pearson correlation analysis was performed between experience (number of ratings) and quality of teaching (average rating). A scatterplot was then graphed, with a quick linear regression model being implemented to add a regression line to the plot.

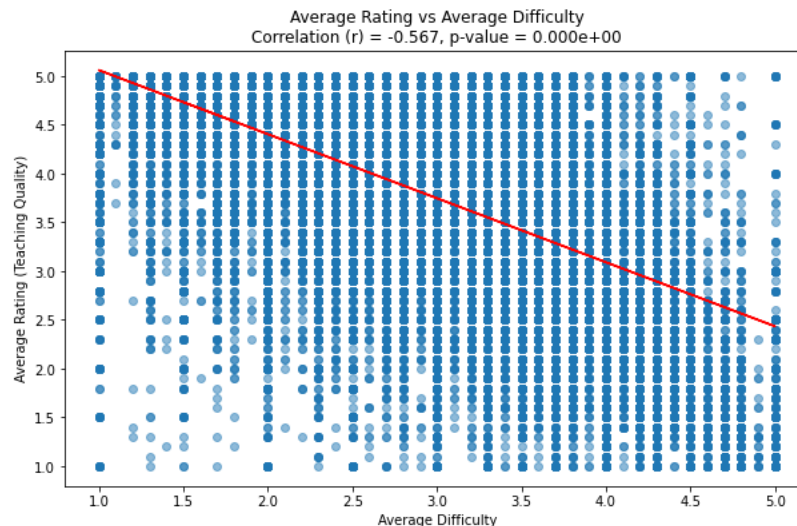
**Results:** A correlation coefficient (**r**) of **0.0402** was obtained, suggesting a very weak linear relationship between experience and the quality of teaching. Further extrapolating from this value, we can tell that we have a **r<sup>2</sup> value** of **0.0016**, indicating that only 0.16% of the variance in the average rating can be explained by the number of ratings. However, the model resulted in an extremely small **p-value** of **3.16e-20**, suggesting that the relationship between the predictor and outcome is statistically significant. This value may have been obtained for reasons other than practical significance, and hence we can ultimately conclude that despite the statistically significant relationship, the weak correlation value suggests that there is no practically meaningful effect of experience on the quality of teaching.



## 3. What is the relationship between average rating and average difficulty?

To determine the relationship between average rating and average difficulty, another Pearson correlation analysis was performed along with the graphing of a scatterplot with a regression line, since these are continuous variables.

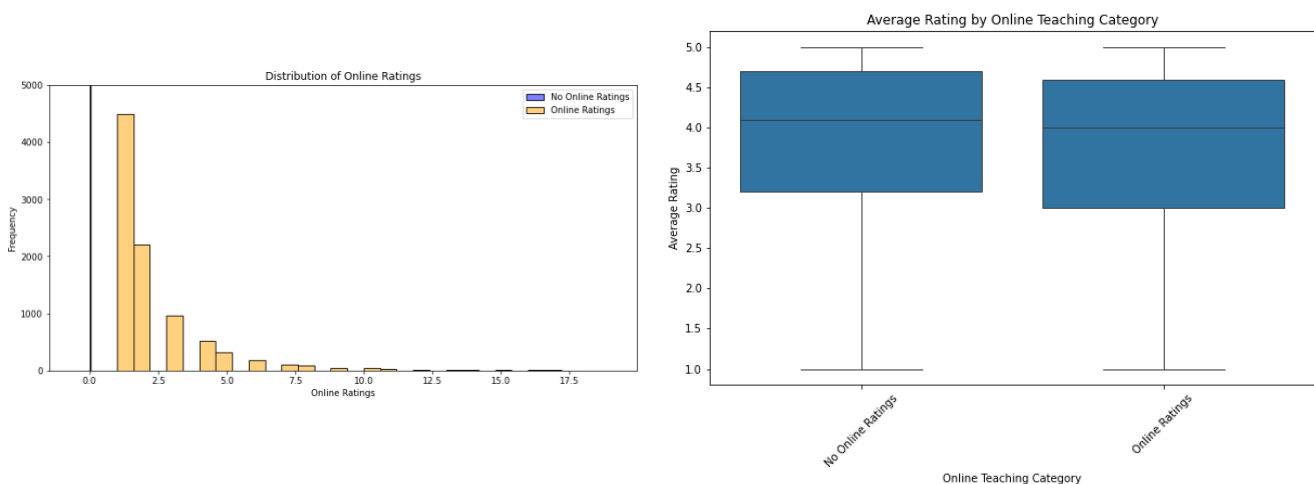
**Results:** A Pearson correlation coefficient (**r**) of **-0.567** was achieved which suggests a moderate to strong negative relationship between average difficulty and average rating. Furthermore, a **p-value** of **0.0** was achieved, suggesting that this correlation is statistically significant at a significance level of 0.005, and this relationship is not due to chance alone. However, an **r<sup>2</sup> value** of **0.321** was obtained, indicating that 32.1% of the variation in average ratings can be explained by average difficulty. This means that despite a significant relationship, there are other factors influencing average rating aside from average difficulty. Ultimately, we can say there is a significant, moderative negative relationship and that as courses become more difficult, students generally provide lower ratings for teacher quality.



4. Do professors who teach a lot of classes in the online modality receive higher or lower ratings than those who don't?

First, some quick data exploration and descriptive statistics of online ratings revealed that most of the number of online ratings are low (with a mean of approximately **0.39**), which a majority of observations recording no online ratings. A histogram was also plotted to visualize the distribution of online ratings, which showed a skewed distribution for professors with online ratings, and a large proportion of professors with no online ratings. Hence, professors were categorized into two groups: those with online ratings of 0, and those with ratings greater than 0. Then a t-test was performed to compare the means of the two groups, with a boxplot graphed to further visualize the difference in ratings between the two categories.

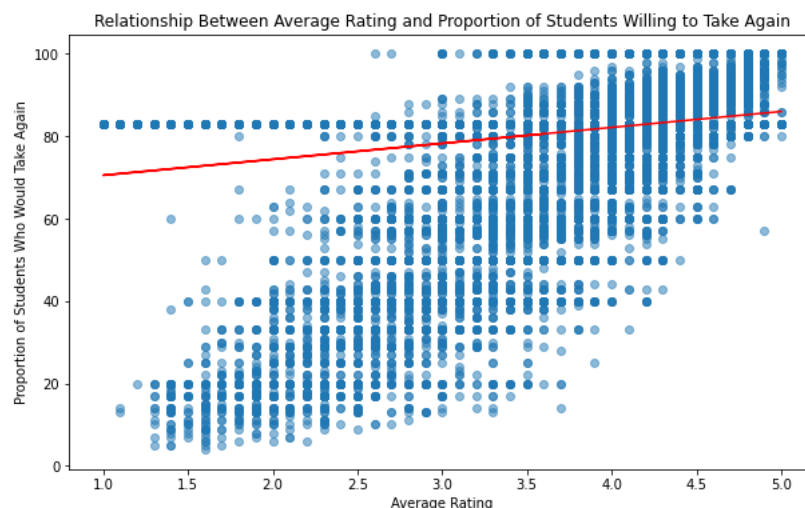
**Results:** A **t-statistic** value of **10.5612** indicates a strong relationship between online teaching modality and average ratings, and that professors who teach more online classes generally receive lower average ratings compared to those who don't. This is further corroborated with a **p-value** of 0.000, and the differences observed in the boxplot, that there is a statistically significant difference between the groups regarding their average ratings.



5. What is the relationship between the average rating and the proportion of people who would take the class the professor teaches again?

To determine the relationship between average rating and the proportion of people who would take the class the professor teaches again, a Pearson correlation analysis was performed, as well as the graphing of a scatterplot with a regression line, since these are continuous variables.

**Results:** A correlation coefficient (**r**) of **0.324** indicates a moderate positive relationship between the average rating and the proportion of people who would take the class the professor teaches again. Furthermore, a **p-value** of **0.00** suggests that this correlation is statistically significant, i.e. the observed relationship between these variables is unlikely to have occurred by chance. However, an **r<sup>2</sup>** value of **0.105** indicates that approximately 10.5% of the variability in students' likelihood to take the class again can be explained by the variability in average ratings. This suggests that there are other factors that may influence students' decisions in retaking a professor's class. In summary, we can conclude that as average ratings increase, the likelihood of students wanting to take the class again also tends to increase.

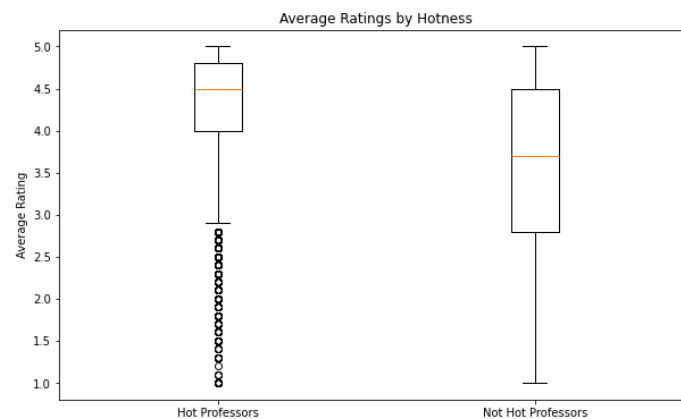


6. Do professors who are “hot” receive higher ratings than those who are not?

Since we are comparing two groups, professors who are considered “hot” and those who are not, professors were categorized into two groups based on the pepper variable: those who were labelled as “hot” (pepper = 1) and those who were not (pepper = 0). Then a t-test was performed, with a boxplot graphed to further visualize the difference in ratings between the two categories.

**Results:** We obtained a very large **t-statistic** of **93.6131**, indicating strong evidence against the null hypothesis, i.e., we can conclude there is a significant difference in average ratings between the "hot" professors and those who are not. Furthermore, a **p-value** of 0.0000 was obtained (p-value < 0.005), indicating that this difference is statistically significant. Hence,

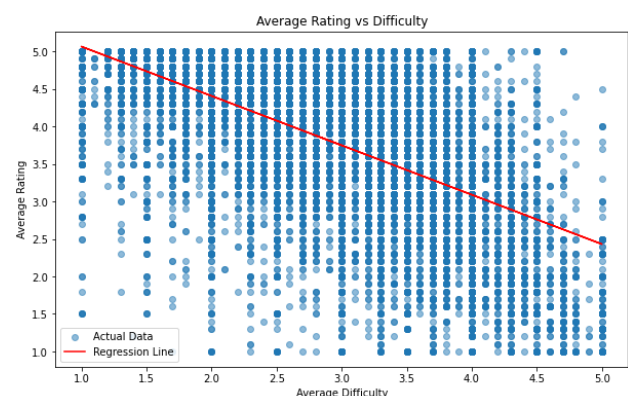
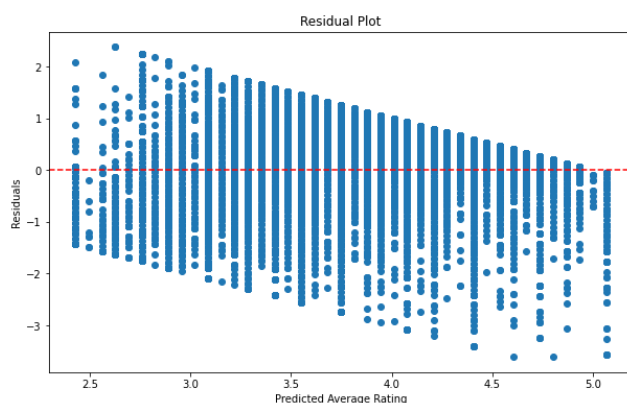
we can conclude that professors who are categorized as "hot" receive significantly higher average ratings than those who are not.



### 7. Build a regression model predicting average rating from difficulty

Using average rating as the predictor and the average difficulty as the outcome, the dataset was split into training and testing sets, with 70% used for training and 30% for testing (because of the relatively small size of dataset) with RNG seeded in 15070423 (my N-number). Then, a simple linear regression model was used to predict average rating from average difficulty, with the model being trained on the training dataset, and then used to predict the ratings in the test dataset. Lastly, a scatterplot of average rating versus average difficulty was graphed with the predicted ratings displayed as a red regression line. Residual Analysis was also conducted with a residual plot to compare the residuals against the predicted values.

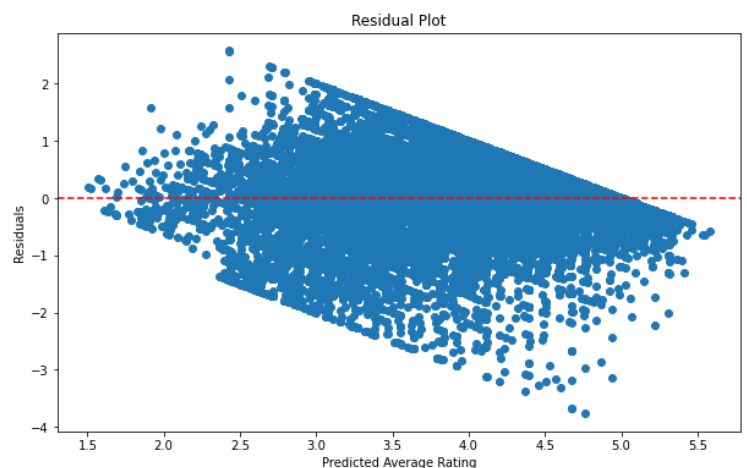
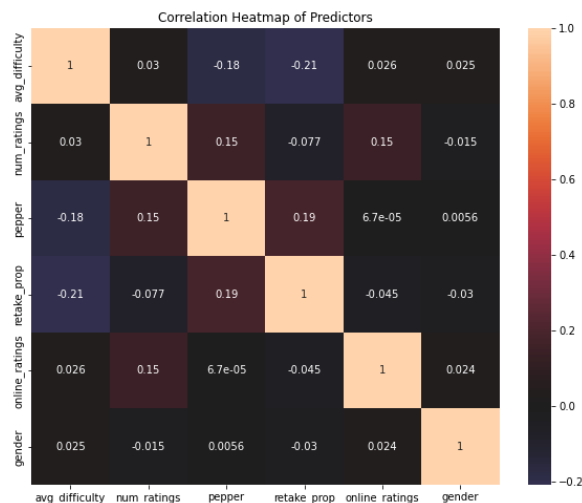
**Results:** The model obtained an **intercept** value of **5.7225**, providing a baseline rating. The **coefficient** or **slope** was found to be **-0.6585** suggesting that as the difficulty of courses increases, students tend to rate the professor lower. Now, for evaluating the model, we obtained a  $r^2$  value of **0.3177** and a root mean squared error (**RMSE**) value of **0.8533** which indicates that the model explains about 31.77% of the variable in average ratings based on the difficulty level of the courses and has an average magnitude of 0.8533 points for prediction errors. This indicates that the relationship between average difficulty and average ratings is statistically significant but not as strong, with the model providing a moderate level of prediction accuracy which can perhaps be improved with the inclusion of other factors.



## 8. Build a regression model predicting average rating from all available factors

Six variables were selected as predictors for this model, with a new column created for gender to avoid multicollinearity issues between the male and female columns. A heatmap was also graphed to display the correlation among the predictors to identify relationships and possible multicollinearity among variables. Then, the features were standardized using StandardScaler to bring them to a common scale in order for linear regression to work effectively. Then, a similar linear regression model was created as the previous question, with a 70-30 split for training and testing, and the graphing of a residual plot to compare the residuals against the predicted values. Lastly, individual betas or regression coefficients for each of the predictors was calculated for analysis.

**Results:** The correlation matrix shows low correlation among any of the variables, eliminating the need for PCA or exclusion of certain variables. With the model, we now achieved a  $r^2$  value of **0.4346** and a root mean squared error (RMSE) value of **0.7802**, indicating that we can now explain a much higher level of variance in average ratings and have smaller prediction errors. This means that adding more predictors has improved the explanatory power of the model and has made it better at predicting average ratings. Upon analyzing the individual betas, we can see that average difficulty is the strongest predictor (**-0.503**) which aligns with the previous question where difficulty along was the focus. Other relatively strong predictors were the perceived “hotness” or pepper (**0.2604**) and retake proportion (**0.18**). On the other hand, the other predictors (number of ratings, gender, and online ratings) had small effects on average ratings.



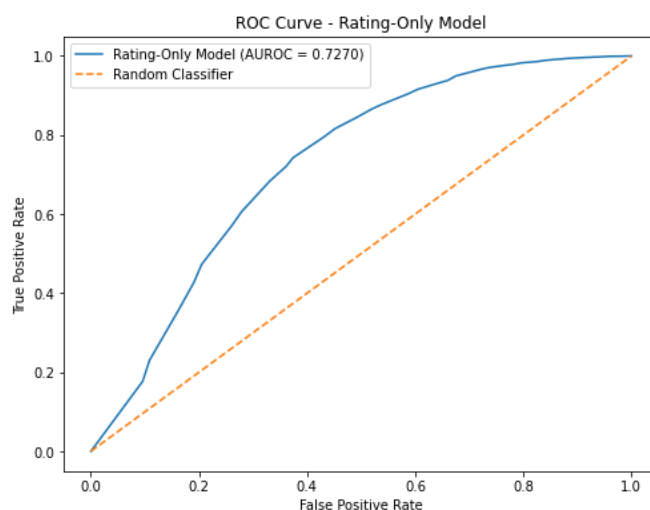
```
Regression Coefficients:
      Predictor      Beta
0  avg_difficulty -0.502128
2      pepper     0.260445
3  retake_prop    0.181451
5      gender    -0.042588
1   num_ratings   0.033726
4  online_ratings -0.021171
```

9. Build a classification model that predicts whether a professor receives a “pepper” from average rating.

Using average rating as the predictor and pepper as a binary outcome (hot or not), data was split into training (70%) and testing (30%) sets, stratified by the target variable (pepper) to ensure balanced representation. Then, a logistic regression model was utilized (for binary classification), with class weights being balanced to handle the imbalance in "pepper" status across the dataset. Finally, the following evaluation metrics were calculated: accuracy, precision, AUROC, confusion matrix, and classification report.

### Results:

The confusion matrix indicated that the model mostly struggles with precision for the “pepper” class as it displayed high number of false positives, but performed better in recall for this class. Furthermore, the classification report and other calculations resulted in an overall **accuracy** of **0.6657**, precision of **0.51**, **recall** of **0.74**, and **F1-score** of **0.60**. This shows that the model displayed moderate performance for the pepper class, with 66.57% of predictions being correct. However, this metric can be misleading due to class imbalance, which is why we turn to AUROC. The model achieved an **AUROC** score of **0.7270**, indicating a fairly good ability to distinguish between the two class (hot vs not hot). This is also indicated by the ROC curve, which is well above the diagonal line, showing that the model is performing better than random guessing.



Confusion Matrix:  
[[6492 3877]  
[1376 3967]]

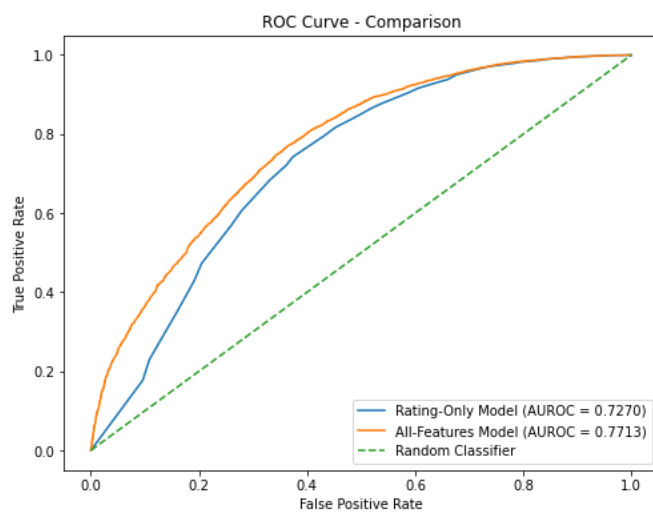
10. Build a classification model that predicts whether a professor receives a “pepper” from all available factors.

Feature selection and standardization was performed in a similar fashion to question 8, while train-test split and the utilization of a logistic regression model along with its evaluation metrics was done similarly to question 9. However, feature importance was calculated for each of the predictors to assess which variables are the most influential in this model.

**Results:** This model as an **accuracy** of **68%**, **precision** of **52%**, and **recall** of **77%**, which are only slight improvements over the rating-only model. However, the AUROC for this



model achieves **0.7713**, which is an improvement over the rating-only model with a AUROC score of 0.7270. Hence, we can say that the model's had improved in its ability to distinguish between classes, even though there are only slight improvements in other evaluation metrics. For the features, we can see that average rating is the most influential predictor (**1.12**), aligning with the strong performance of the rating-only model. Number of ratings and retake proportion contribute to this model's improved performance, though to a less degree, while gender and online ratings have the least influence. Hence, while this model provides a better prediction, the gain is not substantial, suggesting that the average rating alone is a strong predictor.



| Feature Importance: |                |            |
|---------------------|----------------|------------|
|                     | feature        | importance |
| 0                   | avg_rating     | 1.120389   |
| 2                   | num_ratings    | 0.503905   |
| 3                   | retake_prop    | 0.213997   |
| 1                   | avg_difficulty | 0.114328   |
| 5                   | gender         | 0.078902   |
| 4                   | online_ratings | 0.049364   |

**Extra Credit:** Is there a significant difference in average ratings across majors?

For this, the numeric and qualitative datasets were first combined, and then was grouped by major to calculate mean ratings. Majors with very few ratings (<10) were then filtered out to avoid skew. A one-way ANOVA test was then performed (three or more variables) to determine f-statistic, p-value and effect size (which was later calculated). A boxplot was then graphed to represent this difference in ratings

**Results:** We obtained a **F-statistic** of **6.758**, which indicates significant variation in average ratings across majors. We also got a **p-value** of **1.38e-206** indicating that this is a highly statistically significant result, rejecting the null hypothesis that average ratings are equal across all majors, though further analysis should be done on this extremely low p-value. However, while statistically significant, the effect size is small, indicating that differences between majors explain only **3.26%** of the variation in average ratings.



Figure 1 is a plot showing the average rating of 100 majors. The y-axis is labeled 'Average Rating' and ranges from 1.0 to 5.0. The x-axis is labeled 'Major' and lists 100 majors, rotated 45 degrees for readability. Each major has a vertical line with a box plot and a central dot representing the average rating. The ratings are generally distributed between 1.0 and 5.0, with many majors having ratings between 3.0 and 4.0.

Major